

Notes: Zacchia (RES, 2020)

Knowledge Spillovers through Networks of Scientists

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1 Big picture

- **Question.** Do knowledge spillovers across firms actually travel through person-to-person research links between inventors, and if so how large are the resulting social returns to R&D?
- **Main idea.** Instead of measuring spillovers from coarse technological proximity alone, the paper builds a *firm network of scientists*. Two firms are connected when their inventors have former patent collaborators in both organizations.
- **Core empirical object.** For each firm, the paper constructs a connection-weighted external R&D stock,

$$\sum_{j \neq i} g_{ijt} \log S_{jt},$$

where g_{ijt} captures the intensity of inventor-to-inventor cross-firm links.

- **Why this is hard.** If connected firms are hit by common technological or cost shocks, then the R&D of neighbors is endogenous. In addition, network links may themselves be endogenous if better inventors sort toward better firms.
- **Main theoretical contribution.** The paper develops a network game of R&D investment in which shocks are spatially correlated and firms may partially internalize neighbors' conditions. The model implies that R&D becomes independent beyond a sufficiently large network distance.
- **Main identification contribution.** The author instruments the R&D of direct connections with the R&D of *indirect* connections that are sufficiently far away in network space.
- **Main empirical finding.** Connection-weighted external R&D raises firm productivity, market value, and innovation output. The preferred IV and System-GMM estimates imply a spillover elasticity around $\delta \approx 0.021$.
- **Economic interpretation.** The IV estimates are materially larger than OLS estimates, which suggests that naive regressions understate true spillovers because connected firms share confounding forces.
- **Bottom line.** Among firms connected to the network, the marginal social return to R&D is about **112% of the marginal private return**.
- **Why we care.** The paper gives direct evidence that cross-firm human relationships are a transmission channel for industrial knowledge spillovers, and it provides a new network-based IV strategy for spatially correlated settings.

2 Data and measurement (key objects)

2.1 Inventor links and firm-to-firm connections

- The micro-level building block is a co-patenting relationship $p_{mn,t} = 1$ between inventors m and n when, by time t , they have jointly filed at least one patent application in the past.
- Inventors are assigned to firms using the timing of patents filed by that inventor and assigned to the firm. The goal is to infer when an inventor is actively part of a firm's R&D organization.
- A generic firm-to-firm connection measure is

$$c_{ijt}^f = f\left(\frac{\# \text{ inventors of } i \text{ connected to } j + \# \text{ inventors of } j \text{ connected to } i}{\# \text{ inventors of } i + \# \text{ inventors of } j}\right), \quad (3.9)$$

with $f : [0, 1] \rightarrow [0, 1]$, $f(0) = 0$, and $f(1) = 1$.

- The paper’s preferred measure applies the square-root transformation:

$$g_{ijt} = \sqrt{\frac{\# \text{ inventors of } i \text{ connected to } j + \# \text{ inventors of } j \text{ connected to } i}{\# \text{ inventors of } i + \# \text{ inventors of } j}}. \quad (3.10)$$

- **Interpretation.** The square root gives relatively more weight to sparse links. Economically, the author wants to allow even a small number of “insiders” or former collaborators to transmit valuable knowledge.
- **Key properties.**
 - $g_{ijt} \in [0, 1]$ and $g_{iit} = 0$.
 - The measure is symmetric: $g_{ijt} = g_{jit}$.
 - It changes over time because inventors move across firms, collaborate on new patents, enter the sample, or exit the network.

2.2 Main dataset and sample construction

- The firm-level panel comes from Bloom, Schankerman, and Van Reenen (BSV). It contains **736** mostly manufacturing, R&D-intensive U.S. public firms observed from **1976–2001**.
- The panel combines accounting data from COMPUSTAT, patent data, and BSV’s own spillover measures.
- The paper matches the BSV sample to the inventor-disambiguated patent data of Li et al. (2014).
- Final matched patent-inventor universe used for the network:
 - **1,315,060** patents,
 - **565,019** inventors.
- Connections are computed annually from **1981–2001**, using patents granted since 1976 so that prior collaborations can be observed.
- In total, **460** firms have at least one positive connection in some year.

2.3 Important descriptive facts about the network

- The network becomes denser over time: the number of connected firms rises steadily through the 1980s and 1990s, while average links per connected firm keep increasing until the end of the sample.
- Among connected firms, the average connection strength is

$$\mathbb{E}[g_{ijt}] \approx 0.083,$$

with standard deviation about 0.066.

- The row sum of connections,

$$\bar{g}_{it} = \sum_{j \neq i} g_{ijt},$$

has mean about **0.50** and standard deviation about **0.57** among connected firms.

- The degree distribution is highly skewed. The median connected firm goes from roughly 1 link in the early 1980s to roughly 5 by 2000, while the most connected firms accumulate several dozen links.
- The network contains many more open than closed triads: between 1981 and 2001 there are **160,365** open triads but only **15,623** closed triads.
- This matters for identification because open triads generate many indirect friends at distance 2 or 3, which can be used as candidate instruments.

2.4 Firm-level outcomes and controls

- **Main outcome.** Deflated firm sales are used as the productivity outcome in the production-function specification.
- **Other outcomes.** The paper also studies Tobin’s q and citation-weighted patent output in supplementary analysis.
- **Private R&D.** The internal R&D stock S_{it} is built using the standard perpetual-inventory method with 15% depreciation:

$$S_{it} = R_{i,t-1} + 0.85S_{i,t-1}.$$

- **Conventional inputs.** Capital and labor enter the production function.
- **Other spillover controls.**
 - Jaffe’s technological-proximity spillover measure,
 - BSV’s product-market “business stealing” measure,
 - a geographic spillover measure based on the co-location of inventors across CBSAs.
- **Why these extra controls matter.** The coefficient on the scientist-network spillover variable should not just proxy for technology similarity, product-market rivalry, or geographic clustering.

2.5 What Table 1 says about connected firms

- Firms that ever enter the inventor network are systematically different from firms outside it.
- More connected firms are larger, more R&D intensive, more innovative, and more productive.
- For example, firms in the top quartile of cumulative connectedness have average real sales around \$10.5 billion, compared with about \$0.75 billion for firms outside the network.
- Their citation-weighted patent stocks are also much larger: about 652 in the top connected quartile versus only about 7.5 for firms outside the network.
- This is one reason the paper often re-estimates models on the *network-only* subsample: otherwise, simple correlations may partly reflect selection into being a connected innovative firm.

3 Models and empirical specifications (all equations + interpretation)

3.1 Structural knowledge-capital equation

The model starts with a network of firms (I, G) , where $g_{ij} \in [0, 1]$ measures the intensity of the knowledge link between firms i and j . Firm i 's knowledge capital is

$$\tilde{S}_i = S_i^\gamma \left(\sum_{j=1}^N S_j^{g_{ij}} \right)^\delta. \quad (2.1)$$

- S_i . In-house R&D investment.
- S_j . R&D investment performed by other firms.
- g_{ij} . Connection weight that scales how much firm j 's knowledge reaches firm i .
- γ . Elasticity of internal knowledge production with respect to own R&D.
- δ . Strength of knowledge spillovers from connected firms.
- **Economic intuition.** If g_{ij} is larger, the same external R&D stock matters more for firm i . The functional form also makes R&D a strategic complement.

3.2 Structural production function

$$Y_i(X_{i1}, \dots, X_{iQ}; S_1, \dots, S_N) = \left(\prod_{q=1}^Q X_{iq}^{\beta_q} \right) S_i^\gamma \left(\sum_{j=1}^N S_j^{g_{ij}} \right)^\delta e^{\omega_i}. \quad (2.2)$$

- **Conventional inputs.** X_{iq} are capital, labor, and other standard production inputs.
- ω_i . Productivity shock or technological environment shock.
- **Interpretation.** Output depends on own inputs, own knowledge, and external knowledge imported through the inventor network.

The corresponding profit function is

$$\pi_i = \left(\prod_{q=1}^Q X_{iq}^{\beta_q} \right) S_i^\gamma \left(\sum_{j=1}^N S_j^{g_{ij}} \right)^\delta e^{\omega_i} - \sum_{q=1}^Q \xi_q X_{iq} - e^{\zeta_i} S_i, \quad (2.3)$$

where ζ_i is an R&D cost shock. The model allows both ω_i and ζ_i to be spatially correlated in the network.

3.3 Assumptions that make the IV work

- **Assumption 1 (bounded spatial dependence).** There exists some finite integer C such that if the network distance $d_{ij} > C$, then the productivity shocks and R&D cost shocks of firms i and j are independent.
- **Interpretation.** Similar firms may cluster together in network space, but that dependence cannot extend arbitrarily far.

- **Assumption 2 (bounded information).** There exists some finite integer L such that firms do not observe shocks of firms located farther than distance L .
- **Interpretation.** Managers know their own conditions and perhaps those of close neighbors, but not the shocks of firms that are far away in network space.

3.4 Bayes–Nash equilibrium and centrality

The paper solves a game of simultaneous R&D choice. The equilibrium R&D of firm i takes the form

$$\log S_i^* = \frac{\log \gamma + \sum_{q=1}^Q \beta_q (\log \beta_q - \log \xi_q - \log \gamma)}{1 - \gamma - \sum_{q=1}^Q \beta_q} b_i^*(G; \vartheta) + s_i^*(\mathcal{I}_i, G), \quad (2.4)$$

and the optimal conventional inputs satisfy

$$\log X_{iq}^* = \log S_i^* + \log \beta_q - \log \xi_q - \log \gamma + \zeta_i. \quad (2.5)$$

- $b_i^*(G; \vartheta)$. Katz–Bonacich centrality. Better-connected firms have higher deterministic incentives to invest in R&D.
- ϑ . Attenuation parameter,

$$\vartheta = \frac{\delta}{1 - \gamma - \sum_q \beta_q}.$$

- $s_i^*(\mathcal{I}_i, G)$. Recursive component capturing how firm i forecasts the effect of shocks elsewhere in the network.
- **Takeaway.** The network enters both through a deterministic centrality effect and through strategic responses to shocks transmitted via neighbors.

3.5 Key independence proposition behind the instrument

The central identifying result is

$$\text{Cov}(\omega_i, \log S_j^* \mid d_{ij} > C + L) = 0, \quad (2.6)$$

$$\text{Cov}(\log S_i^*, \log S_j^* \mid d_{ij} > C + 2L) = 0, \quad (2.7)$$

$$\text{Cov}(\log S_i^*, \log S_j^* \mid d_{ij} \leq C + 2L) \geq 0. \quad (2.8)$$

- **Interpretation.** Beyond a large enough degree of separation, one firm’s R&D is independent of another firm’s shocks and eventually independent of the other firm’s R&D.
- **IV implication.** Direct neighbors of firm i live at distance 1. If there are firms at distance $D \in [C + L + 1, C + 2L + 1]$, their R&D can predict neighbors’ R&D without inheriting firm i ’s unobservables.
- **Graph 1 intuition.** The tetrad/open-triad graph shows exactly how “friends of friends” can shift the R&D of direct neighbors while remaining exogenous to firm i .

3.6 Empirical production-function specification

The workhorse panel regression is

$$\log Y_{it} = \alpha_i + \sum_{q=1}^Q \beta_q \log X_{itq} + \gamma \log S_{it} + \delta \sum_{j=1}^N g_{ijt} \log S_{jt} + \tau_t + v_{it}. \quad (4.11)$$

- **Dependent variable.** Deflated sales of firm i in year t .
- **Regressor of interest.** The connection-weighted external R&D stock.
- γ . Private-return elasticity of own R&D.
- δ . Spillover elasticity of neighbors' R&D.
- **Fixed effects.** Firm fixed effects α_i and year fixed effects τ_t .
- **Additional controls.** Industry demand variables, Jaffe technological spillovers, business stealing, and geographic spillovers.

3.7 Instrumental-variables moments

Define

$$h_{ikt}^{(D)} = \mathbf{1}\{d_{ikt} = D\},$$

so $h_{ikt}^{(D)} = 1$ when firm k sits exactly D links away from firm i in year t .

Then the paper's key conditional moment is

$$\mathbb{E}[\log S_{kt} v_{it} \mid h_{ikt}^{(D)} = 1] = 0, \quad (4.12)$$

which implies

$$\mathbb{E}[h_{ikt}^{(D)} \log S_{kt} v_{it}] = 0. \quad (4.13)$$

Aggregating over all firms k at that distance yields the implemented instrument

$$\mathbb{E}\left[\left(\sum_{k=1}^N h_{ikt}^{(D)} \log S_{kt}\right) v_{it}\right] = 0. \quad (4.14)$$

- **Economic interpretation.** Distance- D firms move the R&D of my direct neighbors because they share part of the same network neighborhood, but once D is large enough they should not share my residual unobservables.
- **Why distance 2 and 3?** Figure 5 shows R&D correlation at distance 1, weaker correlation at distance 2, and essentially zero beyond distance 3. So the paper experiments with instruments based on distances 2 and 3.
- **Preferred choice.** The paper argues that distance 3 is the safest because large public firms are multi-product and multi-technology, so remaining common shocks should be negligible that far away in network space.

3.8 Dynamic-panel extension: System GMM

To handle endogeneity of standard inputs as well, the residual follows an AR(1) process:

$$v_{it} = \rho v_{i,t-1} + \varepsilon_{it}, \quad \rho \in (0, 1). \quad (5.15)$$

The author then uses System GMM and replaces the standard level-equation instrument for spillovers with a network-distance moment:

$$\mathbb{E} \left[\left(\frac{\sum_{k=1}^N h_{ikt}^{(D)} \log S_{kt}}{\sum_{k=1}^N h_{ik,t-1}^{(D)} \log S_{k,t-1}} \right) \varepsilon_{it} \right] = 0. \quad (5.16)$$

- **Interpretation.** This is the dynamic-panel analogue of the spatial IV logic: use sufficiently distant network variation to instrument the endogenous spillover term in the level equation.
- **Why it matters.** Standard GMM instruments can be weak if a firm's connection structure changes a lot over time. The network-distance instrument improves efficiency and pushes the preferred estimates upward toward the IV-2SLS values.

3.9 From elasticities to social returns

Under a homogeneous percentage increase in all firms' R&D stocks, the marginal social and private returns satisfy

$$\text{MSR}_i = (\gamma + \delta \bar{g}_i) \frac{Y_i}{S_i} \geq \gamma \frac{Y_i}{S_i} = \text{MPR}_i. \quad (5.17)$$

- **Interpretation.** The social return exceeds the private return by the term $\delta \bar{g}_i Y_i / S_i$, which is larger for firms that are more strongly connected in the inventor network.
- **Empirical punchline.** Using the preferred estimates, the paper gets an aggregate MPR around 102% and an aggregate MSR around 114%, so the social return is roughly 112% of the private return.

4 Identification and instruments (what makes the estimates credible)

4.1 Baseline identification problem

- The naive regression of firm outcomes on connection-weighted external R&D is threatened by **correlated effects**: connected firms may share unobserved technological opportunities or R&D cost shocks.
- It is also threatened by **network endogeneity**: inventors and firms may sort into links on the basis of productivity or reputation.
- In reduced-form notation, both problems show up as

$$\mathbb{E}[\log S_{jt} v_{it}] \neq 0 \quad \text{and/or} \quad \mathbb{E}[g_{ijt} v_{it}] \neq 0.$$

4.2 Why the indirect-friends instrument is plausible

- The instrument uses R&D of firms that are not direct neighbors of firm i , but that remain closer to firm i 's neighbors than to firm i itself.
- If spatial correlation in shocks dies out after a finite distance, these indirect links inherit enough relevance to predict neighbors' R&D but not enough proximity to share firm i 's unobservables.
- This is analogous to using deeper lags in dynamic-panel GMM: here the author uses *farther spatial lags* instead of farther time lags.

4.3 Why distance 3 is preferred over distance 2

- Figure 5 shows substantial R&D correlation at distance 2, so a distance-2 instrument might still pick up confounding shocks.
- The paper's own results confirm this concern: using only the distance-3 instrument produces larger spillover estimates than using distance 2 or distances 2 and 3 jointly.
- The author interprets this as evidence that the more plausible case is $(C, L) = (2, 0)$: common shocks, not endogenous reflection of shocks, are what mainly generate spatial dependence in R&D.

4.4 Additional design choices that support credibility

- Models are often estimated on the **network-only subsample** so that the coefficient on the network spillover term is not driven by differences between connected and unconnected firms.
- The author clusters standard errors using **Louvain communities** of the network, which is designed to respect spatial dependence patterns better than naive i.i.d. inference.
- Some specifications also include **community-by-year fixed effects**, which absorb common shocks shared inside coarse network neighborhoods.

5 Tables and figures

5.1 Graph 1: A tetrad, or two semi-overlapping open triads

Read:

- This is the paper's visual intuition for the IV strategy. Firm i is linked to j , and j is linked to k ; the key fact is that k can predict j 's R&D without being directly tied to i .
- The graph shows how a firm's "indirect friend" can be relevant for neighbor R&D but exogenous to the focal firm's shocks.
- It also illustrates why open triads are crucial: if every triad were closed, there would be many fewer clean indirect links available for identification.

5.2 Graphs 2 and 3: Inventor-network examples at $t = 0$ and $t = 1$

Read:

- These graphs explain the connection measure mechanically. A firm-to-firm link is built from inventor-level co-patenting relationships that span organizational boundaries.
- Graph 3 shows how links evolve as inventors move across firms, enter or exit the network, and add new co-patenting ties.
- The main takeaway is that the firm network is genuinely dynamic, not a static technological-similarity matrix.

5.3 Figure 1: Firms, connected firms, and average connections over time

Read:

- The count of connected firms rises strongly from the early 1980s to the late 1990s, then falls somewhat with sample attrition.
- Average links per connected firm rise steadily and sharply, reaching roughly 9–10 by 2001.
- This figure establishes that the network becomes denser over time, which helps both measurement and identification.

5.4 Figures 2 and 3: Degree distribution and distribution of connection strengths

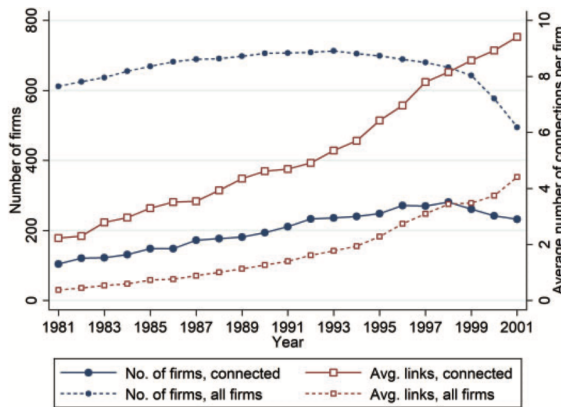


FIGURE 1
Firms, connected firms, and average connections over time

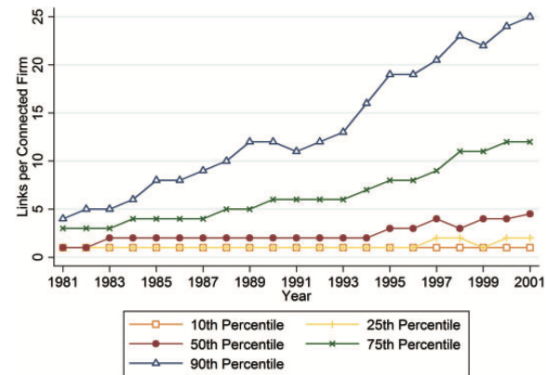


FIGURE 2
Degree distribution over time (connected firms)

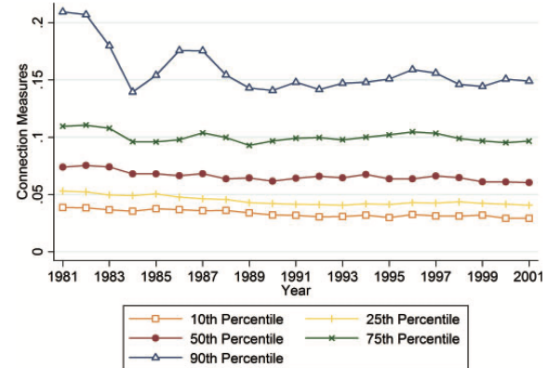


FIGURE 3
Distribution of the connections (g_{ij}) over time

Read:

- Figure 2 shows a strongly skewed degree distribution: the most connected firms accumulate many links, while the median connected firm remains much less connected.
- Figure 3 shows that connection strengths themselves are asymmetric but relatively stable over time. The paper is not finding one or two giant edges dominating the whole network.
- Together, the two figures say that the network gets denser mainly by accumulating more links, not by radically changing the typical strength of each link.

5.5 Figure 4: Triad census

Read:

- Open triads outnumber closed triads by roughly ten to one over the sample.
- This is exactly what the IV strategy wants: plenty of indirect links and not too much clustering.
- If the network were much more closed, the paper would struggle to find firms at useful distances 2 and 3.

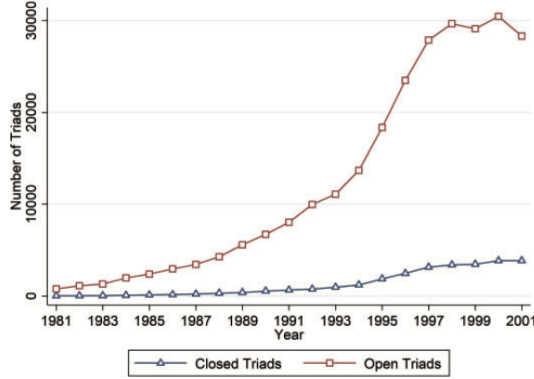


FIGURE 4
Triad census

	No network	1	Quartile of $\sum_i \tilde{g}_i$ 2	3	4
Y_{it} : real sales (Millions 1996\$)	751 (3,792)	1,066 (2,357)	1,383 (2,504)	2,172 (4,533)	10,462 (20,058)
V_{it}/A_{it} : Tobin's q	1.886 (2.031)	1.885 (1.839)	2.573 (3.080)	2.734 (3.306)	3.410 (4.118)
P_{it} : patent stock (cit. weighted)	7.453 (48.17)	16.09 (44.75)	24.65 (50.91)	74.03 (143.8)	652.0 (1322.1)
E_{it} : employees (thousands)	4.068 (12.52)	6.940 (15.80)	9.328 (16.63)	12.40 (22.43)	57.09 (96.80)
Y_{it}/E_{it} : labour productivity	135.6 (80.06)	134.5 (106.6)	157.1 (95.43)	156.5 (117.7)	192.4 (153.3)
Y_{it}/K_{it} : capital productivity	6.932 (6.083)	5.308 (3.167)	5.142 (3.992)	4.941 (3.292)	4.184 (2.883)
Y_{it}/S_{it} : productivity of R&D	39.31 (134.1)	19.71 (70.47)	51.10 (479.9)	11.12 (34.46)	4.342 (3.932)
Y_{it}/J_{it} : Jaffe measure (i, t)	80.28 (407.7)	107.7 (238.5)	140.0 (264.9)	211.6 (435.4)	962.5 (1787.8)
$Y_{it}/\prod_j S_{jt}^{k_{ij}}$: Y to spillover pool		953.9 (2,224.0)	846.2 (1,762.1)	577.6 (1,858.7)	198.9 (1,339.6)
No. of observations	4,363	1,854	1,819	1,949	2,028

Notes: The table is divided in five columns: one for firms in the BSV sample that are never part of the network, and four for each quartile of $\tilde{g}_i$. Standard deviations are in parentheses. S_{it} is the R&D stock, see Section 4. For a description of the Jaffe measure, see [Supplementary Appendix C](#).

5.6 Table 1: Summary statistics, 1981–2001

Read:

- The table splits firms by cumulative connectedness. Firms inside the network are bigger, more innovative, and more productive than firms outside it.
- The top connected quartile has far higher sales, Tobin's q , patent stock, and employment than either the bottom connected quartile or the non-network firms.
- This table motivates why the network subsample is the right sample for the main causal comparison.

5.7 Figure 5: Spatial correlogram of R&D measures

Read:

- Moran's I is very high at distance 1, much smaller but still positive at distance 2, and essentially zero from distance 3 onward.

- This is the key empirical regularity that makes the indirect-friends IV plausible.
- The fact that R&D stock and R&D flow tell the same qualitative story makes the distance-based identification argument much stronger.

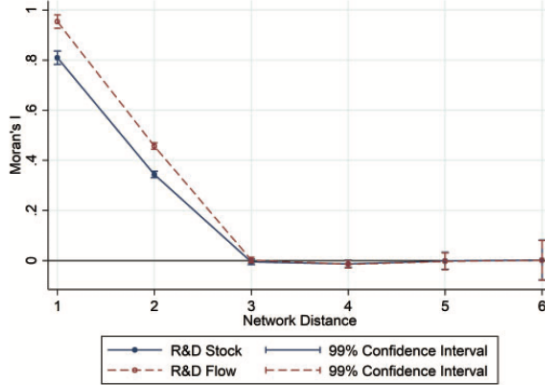


FIGURE 5
Spatial correlogram of R&D measures, 1981–2001

TABLE 2
Production function, ordinary least squares estimates, 1981–2001

	(1)	(2)	(3)	(4)	(5)
Private R&D (γ)	0.0455*** (0.0108)	0.0438*** (0.0105)	0.0568*** (0.0118)	0.0554*** (0.0128)	0.0515*** (0.0142)
R&D spillovers (δ)	0.0159*** (0.0023)	0.0147*** (0.0024)	0.0114*** (0.0024)	0.0118*** (0.0025)	0.0116*** (0.0028)
Geographic spillovers		0.0035 (0.0021)	0.0027 (0.0020)	0.0023 (0.0023)	0.0015 (0.0019)
Capital	0.2071*** (0.0143)	0.2061*** (0.0145)	0.2035*** (0.0194)	0.2020*** (0.0213)	0.2023*** (0.0198)
Labour	0.6550*** (0.0241)	0.6580*** (0.0249)	0.6634*** (0.0351)	0.6613*** (0.0359)	0.6622*** (0.0363)
Jaffe tech. proximity		0.1352** (0.0581)	0.0361 (0.0583)	0.0026 (0.0766)	0.0179 (0.0805)
Fixed effects	YES	YES	YES	YES	YES
Only network	NO	NO	YES	YES	YES
No. of communities (Community $\times$ Year Effects)	0	0	0	10	20
No. of observations	12,503	12,503	7,607	7,607	7,607

Notes: The table reports OLS estimates of model (4.11). Columns 1 and 2 are estimated over the entire original sample of 736 firms in the time interval 1981–2001. Estimates in columns 3, 4, and 5 restrict the sample to firms with at least one non-zero connection ($g_{ijt} \neq 0$) in any year t ; all observations of these firms are also included for years with no connections. All estimates include firm and year fixed effects. Columns 4 and 5 include additional community-by-year fixed effects, where communities are obtained via the Louvain algorithm with $\varphi = 0.8$ (10 communities) in column 4 and $\varphi = 0.6$ (20 communities) in column 5. Standard errors are clustered by the 20 “communities” obtained via the Louvain algorithm with $\varphi = 0.6$ (small sample corrections are applied). All observations of the same individual firm in different years enter the same cluster. For estimates not restricted to the network, firms outside the network constitute single clusters. Asterisks denote conventional significance levels of t -tests (** $p < 0.05$; *** $p < 0.01$).

5.8 Table 2: Production function, OLS estimates

Read:

- In the full sample, the spillover elasticity is around 0.015–0.016 and strongly significant.
- Once the sample is restricted to firms that ever enter the network, the spillover elasticity falls to roughly 0.011–0.012 but remains highly significant.
- The Jaffe technological-proximity measure loses significance in the network-only sample, while the scientist-network measure survives. That is one of the paper’s most important descriptive victories.
- Geographic spillovers are small and statistically weak throughout.

5.9 Table 3: First-stage estimates

Read:

- Both distance-2 and distance-3 indirect-friends instruments predict the endogenous spillover variable.
- The distance-2 first stage is much stronger than the distance-3 first stage, which is exactly what one would expect because distance-2 firms are closer to the focal firm’s neighbors.
- But relevance is not enough; the paper’s argument is that distance 3 is safer on the exclusion side.
- The first-stage F statistics remain comfortably above standard weak-instrument thresholds.

5.10 Table 4: IV-2SLS estimates

Read:

TABLE 3
Production function, first stage estimates, 1981–2001

	(1)	(2)	(3)	(4)	(5)
Distance 2 instrument	0.0043*** (0.0002)	0.0044*** (0.0002)			
Distance 3 instrument			0.0006*** (0.0001)	0.0005*** (0.0001)	0.0005*** (0.0001)
Private R&D	0.1598*** (0.0517)	0.1774*** (0.0488)	0.5022*** (0.1463)	0.4612*** (0.1436)	0.4357*** (0.1504)
Capital	0.1827 (0.1133)	0.1895 (0.1098)	0.6169** (0.2622)	0.5903** (0.2719)	0.5923** (0.2746)
Labour	−0.0761 (0.0921)	−0.0530 (0.0915)	−0.6482** (0.2412)	−0.6021** (0.2636)	−0.6541** (0.2631)
Jaffe tech. proximity	1.8439*** (0.4808)	1.8224*** (0.5123)	3.3663** (1.5953)	3.2740** (1.3777)	3.3672** (1.4690)
Fixed effects	YES	YES	YES	YES	YES
Only network	YES	YES	YES	YES	YES
No. of communities (Community × Year Effects)	0	0	0	10	20
F-statistic	255.17	219.47	24.92	32.06	19.18
No. of observations	7,607	7,607	7,607	7,607	7,607

Notes: The table reports OLS “first stage” regressions of the spillover variable $\sum_{j \neq i} g_{ijt} \log S_{jt}$ on selected instruments and all other right-hand side variables included in the regressions from Table 2. The sample is restricted to firms with at least one non-zero connection ($g_{ijt} \neq 0$) in any year t ; all observations of these firms are also included for years with no connections. Columns 1 and 2 include, on the right hand side, the distance 2 instrument; columns 2 through 5 include the distance 3 instrument. All estimates include firm and year fixed effects. Columns 4 and 5 include additional community-by-year fixed effects, where communities are obtained via the Louvain algorithm with $\varphi = 0.8$ (10 communities) in column 4 and $\varphi = 0.6$ (20 communities) in column 5. Standard errors are clustered by the 20 “communities” obtained via the Louvain algorithm with $\varphi = 0.6$ (small sample corrections are applied). All observations of the same individual firm in different years enter the same cluster. Asterisks denote conventional significance levels of t -tests (* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$).

TABLE 4
Production function, two stages least squares estimates, 1981–2001

	(1)	(2)	(3)	(4)	(5)
Private R&D (γ)	0.0560*** (0.0117)	0.0562*** (0.0118)	0.0510*** (0.0111)	0.0489*** (0.0127)	0.0464*** (0.0131)
R&D spillovers (δ)	0.0127*** (0.0029)	0.0125*** (0.0030)	0.0204** (0.0084)	0.0230** (0.0088)	0.0211** (0.0095)
Geographic spillovers	0.0027 (0.0019)	0.0027 (0.0019)	0.0030 (0.0018)	0.0026 (0.0021)	0.0018 (0.0018)
Capital	0.2025*** (0.0200)	0.2027*** (0.0199)	0.1969*** (0.0225)	0.1944*** (0.0244)	0.1956*** (0.0234)
Labour	0.6642*** (0.0357)	0.6640*** (0.0356)	0.6692*** (0.0373)	0.6677*** (0.0382)	0.6685*** (0.0392)
Jaffe tech. proximity	0.0314 (0.0549)	0.0324 (0.0553)	0.0041 (0.0545)	−0.0364 (0.0669)	−0.0167 (0.0744)
Spillovers IV(s)	$D = 2$	$D = 2, 3$	$D = 3$	$D = 3$	$D = 3$
Fixed effects	YES	YES	YES	YES	YES
Only network	YES	YES	YES	YES	YES
No. of communities (Community × Year Effects)	0	0	0	10	20
No. of observations	7,607	7,607	7,607	7,607	7,607

Notes: The table reports IV-2SLS estimates of model (4.11). All estimates are restricted to firms with at least one non-zero connection ($g_{ijt} \neq 0$) in any year t ; all observations of these firms are also included for years with no connections. Models in columns 1 and 2 employ the distance 2 instrument; models in columns 2 through 5 employ the distance 3 instrument. All estimates include firm and year fixed effects. Columns 4 and 5 include additional community-by-year fixed effects, where communities are obtained via the Louvain algorithm with $\varphi = 0.8$ (10 communities) in column 4 and $\varphi = 0.6$ (20 communities) in column 5. Standard errors are clustered by the 20 “communities” obtained via the Louvain algorithm with $\varphi = 0.6$ (small sample corrections are applied). All observations of the same individual firm in different years enter the same cluster. Asterisks denote conventional significance levels of t -tests (* $p < 0.05$; ** $p < 0.01$).

TABLE 5
Production function, system GMM estimates, 1981–2001

	(1)	(2)	(3)	(4)	(5)
Private R&D (γ)	0.1234** (0.0470)	0.0786** (0.0359)	0.0959** (0.0367)	0.0880*** (0.0306)	0.0668* (0.0366)
R&D spillovers (δ)	0.0098 (0.0099)	0.0142 (0.0092)	0.0167** (0.0092)	0.0212*** (0.0088)	0.0207*** (0.0087)
Geographic spillovers	0.0160 (0.0136)	0.0189* (0.0104)	0.0067 (0.0123)	0.0036 (0.0124)	0.0063 (0.0113)
Capital	0.2731*** (0.0806)	0.2580** (0.0961)	0.2603** (0.0912)	0.2743*** (0.0936)	0.2947** (0.1054)
Labour	0.5555*** (0.0836)	0.6096*** (0.0990)	0.6136*** (0.0934)	0.5992*** (0.1036)	0.6049*** (0.1172)
Jaffe tech. proximity	−0.0555* (0.0313)	−0.0391 (0.0277)	−0.0590** (0.0249)	−0.0783*** (0.0247)	−0.0682** (0.0281)
Lags s of GMM-style IVs	$s \geq 3$	Standard	Standard	Standard	Standard
Diff. eq. spillover IVs	Standard	$D = 2$	$D = 3$	$D = 3$	$D = 3$
Level eq. spillover IVs	Standard	$D = 2$	$D = 3$	$D = 3$	$D = 3$
Time effects	YES	YES	YES	YES	YES
Only network	YES	YES	YES	YES	YES
No. of communities (Community × Year Effects)	0	0	0	10	20
No. of instruments	978	962	962	1,142	1,342
No. of observations	7,185	7,185	7,185	7,185	7,185

Notes: The table reports System GMM estimates of model (4.11). All estimates are restricted to firms with at least one non-zero connection ($g_{ijt} \neq 0$) in any year t ; all observations of these firms are also included for years with no connections. The ρ -differenced production function is estimated by one step System GMM: lags s of the endogenous variables in levels are employed as instruments in the differenced equation; lags s of the same variables in first differences are used as instruments in the level equation. In columns 2 through 5 the standard GMM instruments of the spillover variables for the level equation are substituted by moments (5.16) for the given value of D . All estimates include year fixed effects. Columns 4 and 5 include additional community-by-year fixed effects, where communities are obtained via the Louvain algorithm with $\varphi = 0.8$ (10 communities) in column 4 and $\varphi = 0.6$ (20 communities) in column 5. Structural parameters are recovered from the steady-state representation of the ρ -differenced model; standard errors are calculated by the Delta Method (see [Supplementary Appendix E](#) for details). Standard errors are clustered by the 20 “communities” obtained via the Louvain algorithm with $\varphi = 0.6$ (small sample corrections are applied). All observations of the same individual firm in different years enter the same cluster. Asterisks denote conventional significance levels of t -tests (* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$).

- Using the distance-2 instrument alone gives $\hat{\delta} \approx 0.0127$, only slightly above OLS.
- Using the distance-3 instrument alone raises the estimate sharply to about 0.020–0.023.
- This is the core causal pattern in the paper: once the instrument is far enough away to avoid confounds, the spillover elasticity nearly doubles relative to network-subsample OLS.
- The result survives the addition of community-by-year fixed effects.

5.11 Table 5: System GMM estimates

Read:

- Standard System GMM without the network-based level-equation instrument gives a small and insignificant spillover estimate, about 0.010.
- Replacing the standard level-equation spillover instrument with the distance-3 network instrument raises the estimate to roughly 0.017–0.021.
- In the preferred columns with community-year effects, $\hat{\delta} \approx 0.021$ and remains statistically

significant.

- This is important because it shows the main result is not an artifact of treating capital, labor, and own R&D as exogenous.

6 Short summary

- **Conclusion.** The paper finds that inventor-to-inventor links across firms are an economically meaningful channel of R&D spillovers.
- **Intuition.** Knowledge moves through people. When firms share inventors with past collaborators in other organizations, outside R&D becomes more productive for them.
- **Empirical lesson.** A network-distance IV can separate real spillovers from correlated shocks, and in this application it suggests that simple OLS understates the true size of R&D spillovers.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Connection-weighted external R&D significantly increases firm productivity.
- **Preferred magnitude.** The best IV and System-GMM estimates imply $\delta \approx 0.021$.
- **Comparison with OLS.** OLS on the network subsample gives only about 0.011–0.012, so causal estimates are almost twice as large.
- **Interpretation.** This pattern is consistent with negative bias in OLS due to common confounders or endogenous network formation.
- **Quantitative implication.** Using the preferred estimates, the aggregate marginal social return to R&D among connected firms is about 114%, versus a private return around 102%.
- **Broader evidence.** Supplementary results for Tobin's q and patent output point in the same direction: the spillover effect survives beyond the production-function outcome.

7.2 Economic intuition

- Firms do not absorb outside knowledge from every technologically similar firm equally. They learn more from firms that are actually connected through scientists.
- A small number of shared inventor relationships can matter a lot because those people carry tacit knowledge, informal information, and awareness of what other labs are doing.
- Because technologically related firms can also face similar cost and opportunity shocks, naive correlations mix spillovers with shared environments. The indirect-friends IV is designed to peel those two apart.
- The fact that distance-3 instruments work best suggests that what matters empirically is not reflection of shocks through strategic interaction alone, but broader exogenous similarity in the network neighborhood.

7.3 Takeaway

This paper upgrades the empirical study of R&D spillovers from “firms in similar technological space” to “firms connected through actual people.” That move matters both conceptually and quantitatively: the network of scientists predicts where spillovers travel, and once the author isolates plausibly exogenous network variation, the social payoff to R&D looks substantially larger than the private one. For anyone studying innovation, agglomeration, or firm networks, the paper’s main lesson is that *who is linked to whom* is not just descriptive detail — it is part of the causal mechanism.